

Udit Agarwal

+1-930-904-4901 | agarwaludit13@gmail.com | linkedin.com/in/udit013 | github.com/Udit013 | uditagarwal.vercel.app

EDUCATION

Indiana University Bloomington

Master of Science in Computer Science, GPA: 3.84/4.0

Bloomington, Indiana, US

Aug 2024 - May 2026

Kalinga Institute of Industrial Technology (KIIT University)

Bachelor of Technology in Computer Science and Engineering, GPA: 8.85/10.0

Bhubaneswar, Odisha, India

Aug 2020 - May 2024

EXPERIENCE

Heartland Community Network

Senior Consultant

Jun 2026 - Present

Remote, Indiana, US

- Design and develop full-stack applications, backend services, and REST APIs tailored to client operational requirements, contributing across the complete development lifecycle from design through production deployment
- Build AI-powered solutions leveraging LLMs, data analytics, and workflow automation to streamline business processes and improve decision-making for clients across multiple industries

Global Health Impact Project

Software Engineer

Indiana University, Bloomington, Indiana

Jan 2026 - Jun 2026

- Built full-stack features and REST APIs for a global health analytics platform (React, TypeScript, Python), enabling researchers to evaluate pharmaceutical intervention effectiveness across populations
- Developed scalable backend services and data-processing pipelines supporting treatment-coverage modeling, efficacy analysis, and large-scale forecasting workflows

University Information Technology Services (UITS)

Consultant - IT Services

Indiana University, Bloomington, Indiana

Aug 2025 - Present

- Resolved 100+ technical issues weekly across desktop, mobile, and enterprise systems using structured troubleshooting and root-cause analysis, documenting resolutions and recurring incidents in ServiceNow
- Supported identity and access management workflows, ensuring secure authentication and account lifecycle management across university systems

PROJECTS

Brain Tumor MRI Classifier - Production Serving (IEEE 2024) | *Python, ONNX Runtime, FastAPI,*

Docker, TensorFlow/Keras, Grad-CAM

- Served a calibrated MRI classifier as an ONNX + FastAPI endpoint returning class, confidence, and Grad-CAM overlays at 33 ms CPU inference, containerized with Docker
- Co-authored the underlying IEEE 2024 publication benchmarking four CNN architectures on 7,000+ scans, with EfficientNetB3 best at just 11.7M parameters
- Audited the dataset with perceptual-hash deduplication to remove train/test leakage and validated honest performance (93.9% accuracy, 0.985 macro ROC-AUC)

Automated Secrets Scanner | *Python, FastAPI, React, TypeScript, PostgreSQL, SQLite, WebSocket,*

APScheduler

- Engineered a secret-detection engine covering 26 credential types using regex matching, Shannon-entropy analysis, and false-positive suppression, reaching 95% precision, 94% recall, and 0.94 F1
- Built an asynchronous FastAPI backend (18 REST endpoints, WebSocket progress, APScheduler scheduling, SMTP alerting) with a 0-100 exposure-risk model and differential scanning, covered by a 75-test CI suite
- Implemented SARIF 2.1.0, JSON, and CSV export for GitHub Advanced Security, plus git-history lifecycle intelligence and a console with hand-built SVG charts (410 KB bundle cut)

AI Voice Interview Simulator | *Next.js, TypeScript, Firebase, Web Speech API, Monaco Editor,*

Pyodide/WASM, Gemini, Zod

- Built a full-stack AI interview platform running end-to-end voice interviews with browser-native speech recognition and synthesis (no paid STT/TTS) across five interview formats
- Engineered an adaptive, delivery-aware engine on Gemini measuring hesitation, pace, and filler density each turn to adjust difficulty, with config-driven company interview modes from one template registry
- Built live coding rounds in a Monaco editor running JavaScript and Python in a sandboxed, terminable Web Worker (Pyodide/WASM), with session auth, Zod-bounded requests, and a 36-test CI suite

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

Frontend: React, Next.js, Tailwind CSS

Backend: FastAPI, Fastify, Node.js, Express.js, REST APIs, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite, pgvector

AI and ML: PyTorch, TensorFlow, scikit-learn, Hugging Face, LLMs (RAG, LangGraph, Fine-Tuning), ONNX

Cloud and DevOps: AWS, Docker, CI/CD, Git, ServiceNow, Postman